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An Improved Description of the Small RNA Landscape of the Human Fungal Pathogen *Aspergillus fumigatus*

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Received: 29 October 2025 | **Revised:** 17 November 2025 | **Accepted:** 13 December 2025

Keywords: double-stranded RNA | filamentous fungi | fungal pathogen | small RNA | tRNA-derived RNAs

ABSTRACT

Aspergillus fumigatus is a ubiquitous filamentous fungus and dangerous human pathogen that produces a limited pool of small RNAs under standard laboratory conditions. To better understand the rules of small RNA production in *A. fumigatus*, we induced canonical RNA interference (RNAi) via overexpression of two separate inverted-repeat transgenes. We observed production of predominantly 20-nt, 5' uridine-containing small RNAs from the 3' end of each transgene nearest to the loop region and dependent on dicer-like B RNase III enzyme. Using this refined knowledge, we assessed small RNA biogenesis by sRNA-seq in three double knockout strains of the RNAi pathway, namely the dicer-like proteins ($\Delta dclA/B$), argonautes ($\Delta pdA/B$), and RNA-dependent RNA polymerases ($\Delta rrpA/B$). In each case, we found limited evidence for production of 5' U-containing small RNAs reliant on the RNAi machinery under standard laboratory conditions. We did observe 5' U-containing small RNAs amongst the abundant tRNA-derived RNAs (tDRs); however, biogenesis of tDRs was predominantly Dicer-like independent. To more accurately define the complex tDR repertoire, we employed a cutting-edge tDR-sequencing approach that improved tRNA-half detection and revealed qualitative morphotype-specific changes in the small RNA fraction of conidia relative to mycelium. Finally, leveraging the limited sRNA repertoire of *A. fumigatus*, we tested the consequences of inverted-repeat transgene overexpression in the $\Delta dclA/B$ double knockout, which revealed growth inhibition even in the absence of double-stranded RNA (dsRNA) processing and small RNA production. We hypothesize that the RNAi substrate-limited landscape of *A. fumigatus* facilitates sensitivity to increases in dsRNA, offering an intriguing system for future studies of dsRNA metabolism.

1 | Introduction

The canonical RNA interference (RNAi) pathway processes double-stranded RNA (dsRNA) through cleavage to small RNAs (sRNA) that are then used as guide molecules to facilitate silencing of target transcripts (Wilson and Doudna 2013). The dsRNA precursor molecule can be produced endogenously via transcription of self-genes or exogenously, for example, from a virus or repetitive transgene. The dsRNA A-form helix

is characterized by a narrowed major groove compared to B-form DNA, meaning that the molecule itself can be a substrate for multiple dsRNA binding proteins (dsRBPs) independent of the nucleotide sequence (Cottrell et al. 2024). The result is competition for dsRNA substrates by dsRBPs that bind, cleave, or edit the dsRNA molecule through interactions with the sugar-phosphate backbone (Cottrell et al. 2024). In humans for example, this manifests as a balance between processing (RNA cleavage or modification) and activation of inflammatory

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cascades by dsRBP sensors like MDA5 (Liddicoat et al. 2015), whereas in microbes like the bacteria *Escherichia coli*, dsRNA is thought to be rapidly degraded or processed to smaller RNAs (Lybecker et al. 2014). In many cases, how fungi process dsRNA remains poorly understood.

The sRNA landscape of fungi (like many eukaryotes) consists of a heterogeneous group of RNA molecules, ranging from fragments of housekeeping RNA like tRNA and rRNA to small interfering RNAs (siRNA) and microRNA-like RNAs (miRNA) produced from both perfectly duplexed dsRNA and dsRNA structures with bulges and/or mismatches (Torres-Martinez and Ruiz-Vazquez 2017). Fungal sRNAs are produced from these precursors via diverse biogenesis pathways, often including members of the canonical RNAi machinery such as the RNase III enzyme Dicer (or Dicer-like), Argonaute, and/or RNA-dependent RNA polymerases (RdRPs), together with peripheral factors (Lee et al. 2010). Some fungi have lost RNAi to gain an advantage in a particular niche (Billmyre et al. 2013), but many, including the ubiquitous human pathogen *Aspergillus fumigatus*, harbor an active RNAi pathway (Mouyna et al. 2004) capable of producing small RNAs in response to mycovirus infection (Ozkan et al. 2017). Interestingly, *A. fumigatus* appears to produce a limited pool of endogenous sRNAs under standard growth conditions, consisting in large part of tRNA-derived RNAs (tDRs; Figure S1) despite encoding two Dicer-like proteins, two Argonautes, and two RdRPs (Table S1; Kelani et al. 2023; Ozkan et al. 2017; Jochl et al. 2008).

The tDRs are an emerging class of heterogeneous 10- to 50-nucleotide (nt) sRNAs that include the 5' and 3' tRNA halves (tRHs; ~31 to 40 nt), tRNA fragments (tRF) of 17 to 25 nt in length processed from the 5' (tRF-5) and 3' ends (tRF-3, tRF-3 CCA), internal tRFs (i-tRF), and other miscellaneous fragments (Tyczewska and Grzywacz 2022) (Figure S1). The generation of tDRs is multifactorial and dynamic across development, relying on enzymes like the RNase A ribonuclease angiogenin, RNase T2-type enzymes like Rny1p in yeast, or even Dicer in some species (Donovan et al. 2017; Ivanov et al. 2011; Jochl et al. 2008; Thompson et al. 2008; Luhtala and Parker 2010; Loveland et al. 2024; Babiarczyk et al. 2008; Cole et al. 2009; Haussecker et al. 2010; Kuscu et al. 2018; Haiser et al. 2008; Thompson and Parker 2009). In *A. fumigatus*, tRHs accumulate after germination from conidia to hyphae independently of mature tRNA levels; however, during early conidiogenesis there is a rapid increase in tRHs and a decrease in mature tRNA levels (Jochl et al. 2008). In that study, it was posited that the decrease in the mature tRNA pool in conidia may contribute to the metabolic dormancy of spores as active translation is shut down. The tRF-3 RNAs are known to interact with the Argonaute-containing RNA-induced silencing complex (RISC) to guide silencing of target mRNA based on 3' UTR seed sequence complementarity (Maute et al. 2013; Thompson and Parker 2009) and to facilitate ribosomal RNA processing in association with Argonaute (Couvillion et al. 2010). Despite the existence of foundational work in this area, the exact small tRNA-derived RNAs produced in conidia versus hyphae remain unknown.

Here, we provide an improved global characterization of the sRNA landscape and dsRNA metabolism of the World Health

Organization critical priority pathogen *A. fumigatus* (Fisher and Denning 2023) using sRNA-seq and tDR-seq approaches. Although the endogenous repertoire of sRNAs is limited, we show that heterologous overexpression of inverted-repeat transgenes results in production of canonical 20-nt 5' U-containing small RNAs and limits fungal growth even in the absence of dicer-like proteins, suggesting that excess dsRNA may itself serve as a check on growth in *A. fumigatus*.

2 | Results

2.1 | The *A. fumigatus* RNAi Machinery Processes Inverted-Repeat Transgene dsRNA to 20-nt, 5' U-Containing sRNAs

We previously showed that overexpression of long dsRNA in *A. fumigatus* as an inverted-repeat transgene (IRT) with complementarity to a nonessential gene (the polyketide synthase *pksP*) results in target gene silencing and surprising growth inhibition in liquid culture (Kelani et al. 2023). To refine our understanding of dsRNA processing in *A. fumigatus*, we employed sRNA-seq of *pksP* IRT-expressing wild-type and dicer-like double knockout strains ($\Delta dclA/B$ -IRT; Kelani et al. 2023) after induced expression of the transgene in liquid culture. We grew the *pksP* IRT strains for 16 h plus 8 h of induction to ensure sufficient biomass prior to collection of RNA and sRNA-seq analysis (Figure S2A,B). This allowed us to assess the ability of *A. fumigatus* to produce sRNAs from the transgene transcript. Untreated wild-type fungus was included as a control for standardization with previous experiments. sRNAs were mapped to the *pksP* IRT sequence, and we observed an accumulation of 20-nt RNAs with a 5' uridine produced from the transgene in the wild-type strain (Figure 1A–C), revealing that the *A. fumigatus* RNAi machinery can function in a manner consistent with the fungal literature (Regmi et al. 2023; Mi et al. 2008). These peaks of sRNA were absent in the wild-type and $\Delta dclA/B$ -IRT strains but did remain when we performed the same experiment in $\Delta ppdA/B$ and $\Delta rrpA/B$ *pksP* IRT expressing strains (Figure S2C–E). Closer inspection of the mapped reads revealed that the largest accumulation originated from the 3' half of the transcript (antisense) primarily as approximately two 20-nt RNAs (Figure 1C, Figure S2D,E). Prediction of target binding sites using these two relatively abundant sRNA sequences revealed no additional high-quality targets beyond the known complementarity to the *pksP* gene, potentially suggesting that the toxicity observed by the expression of the inverted-repeat transgene in our previous study is not facilitated through small RNA off-target effects (Kelani et al. 2023).

sRNA-seq of *pksP* IRT-expressing single knockouts of *dclA* and *dclB* revealed that DclB, but not DclA, is required for production of the observed 5' U-containing sRNAs (Figure 1D). Interestingly, even in this second, separate IRT-sequencing experiment, the pattern of cleavage of the hairpin RNA was nearly identical to that observed in Figure 1C providing corroboration of DclB specificity preferences. As an additional control, we confirmed the activity of DclB through complementation of the $\Delta dclB$ -IRT strain with a functional copy of *dclB* driven by its native promoter (Figure S3). To confirm that the length of 20 nt and a 5' U were general features of DclB

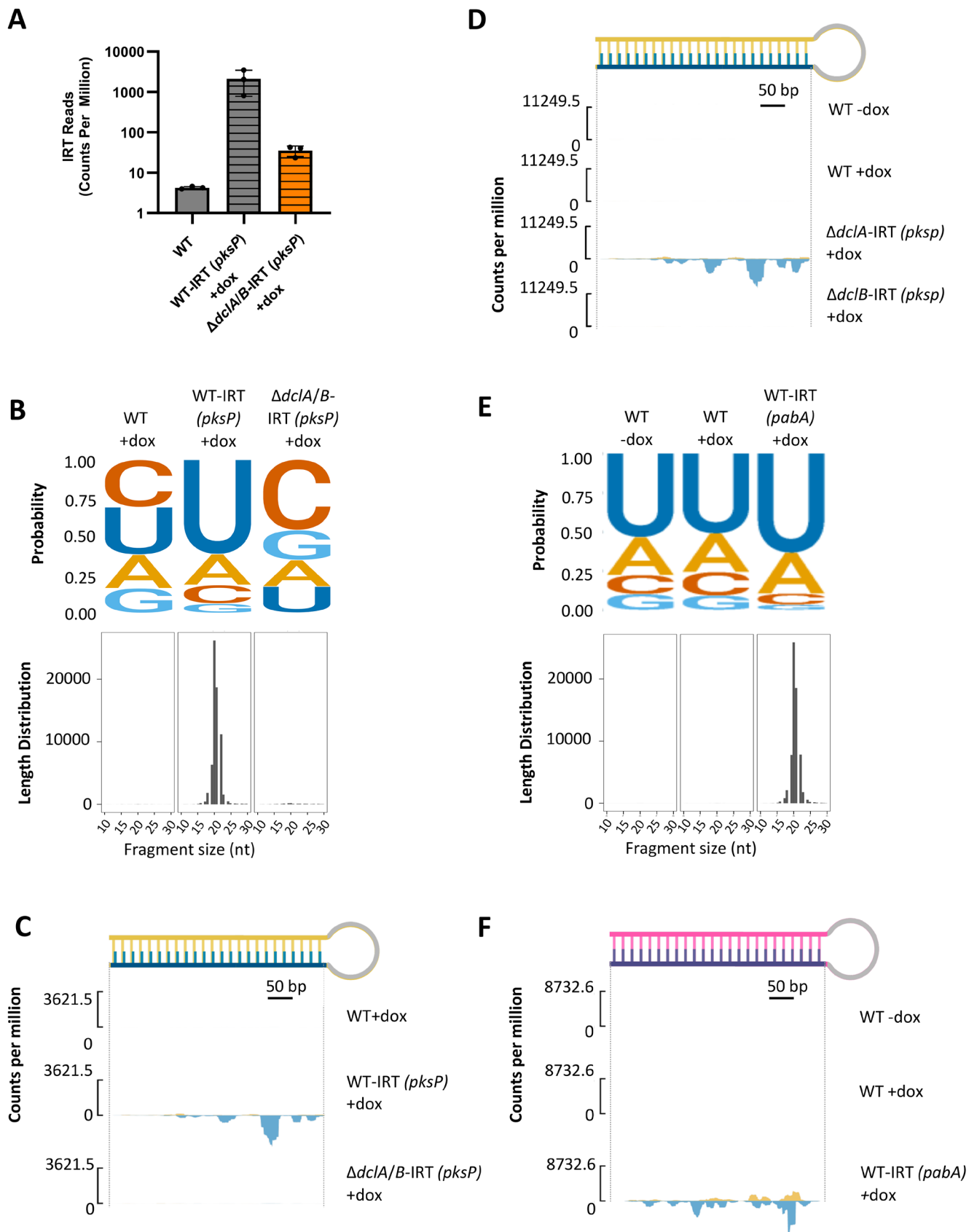


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FIGURE 1 | Inverted-repeat transgene expression induces 20-nt, 5' U-containing sRNA production. (A) sRNA reads mapped to *pksP* inverted-repeat transgene (IRT) sequence per million mapped reads for each of the denoted strains. Data is accumulated from three biological replicate sRNA-seq experiments. (B) The distribution of the 5' nucleotide for the respective strains (top) and histogram showing the average sRNA read length for the sRNAs that mapped to the *pksP* IRT (bottom). (C) Graph showing read coverage of *pksP* inverted-repeat transgene region with reads from the 5' end of the inverted repeat shown as positive strand and from the 3' end shown as negative strand. (D) Graph showing read coverage of *pksP* inverted-repeat transgene region in the $\Delta dclA$ and $\Delta dclB$ knockout strains. (E) The distribution of the 5' nucleotide for the respective strains (top) and histogram showing the average sRNA read length for the sRNAs that mapped to the *pabA* IRT (bottom). (F) Graph showing read coverage of *pabA* inverted-repeat transgene region in the respective strains. Graphs in (C, D, F) were created with the SparK package (Kurtenbach and William Harbour 2019) with reads from the 5' end of the inverted repeat shown as positive strand and from the 3' end shown as negative strand.

processing of hairpin RNAs, we performed additional sequencing of a *pabA* IRT-expressing wild-type strain that was created previously (Kelani et al. 2023), and again observed 20-nt, 5' U-containing sRNAs primarily produced from the 3' end of the transcript (Figure 1E,F). Additional biochemical experiments in vitro with purified enzymes and defined substrates will be required to fully elucidate the mechanism of DclB cleavage, but these results already suggest some DclB preferences in *A. fumigatus*.

2.2 | The Composition of *A. fumigatus* Endogenous sRNAs Is Morphotype Specific

After improving our understanding of small RNA biogenesis, we next employed previously described knockout strains in the canonical RNA interference machinery ($\Delta dclA/B$, $\Delta ppdA/B$, and $\Delta rrpA/B$) created in the wild-type CEA17 $\Delta akuB^{KU80}$ background to simultaneously profile expression of small noncoding RNA (ncRNA; this study) and mRNA (Kelani et al. 2023) from the same biological samples under normal growth conditions (conidia, 24-h mycelium, and 48-h mycelium; Figure 2A; Figure S4A; Data S1). We mapped the expression of distinct small ncRNA classes, including snRNA, snoRNA, and tDRs (Figure 2B). In conidia, many reads mapped to rRNA, whereas the proportion of tDRs and other small ncRNAs increased in the mycelial samples, consistent with an expected, more diverse transcriptome in metabolically active mycelium. Across all the sequencing samples, rRNA made up the majority of the mapped reads, followed by short reads mapping to tRNA genes. An investigation of the 19–25 nt small RNAs derived from rRNA revealed no 5' nucleotide sequence bias or change after deletion of RNAi components (Figure S4B).

We next aimed to identify sRNAs that might be masked by the high abundance of rRNA reads. Removal of reads matching rRNA and restriction of read length to 18 to 30 nt led to identification of similar numbers of sRNAs from each of the double knockout strains compared to wild type (Figure S4C,D). In conidia, a clear population of 18 to 21 nucleotides was identified, whereas in 24- and 48-h mycelium, we generally observed two or more peaks centered at ~19 nt and/or ~28 nt, often indicative of tRNA-derived RNAs. We then used Manatee (Handzlik et al. 2020) to identify and predict sRNAs, including microRNA-like RNAs (miRNA; Data S2). We found no sequence bias in the 19–25 nt small RNA fraction from any condition (Figure 2C). We observed few low abundance candidate sRNAs in conidia, and a larger number of slightly higher abundance sRNAs from 24- and 48-h mycelium. Interestingly, deletion of *dclA/B* did not change

the overall number of candidate sRNAs identified (1524 compared to 1615 in wild type), whereas loss of *ppdA/B* and *rrpA/B* decreased the number of sRNAs identified by Manatee (873 and 1016 respectively), although most of the candidates identified were not of particularly high abundance (Data S2). Intersection of the Manatee-identified sRNAs using UpSetR showed limited overlap between the different strains (Figure S5A), with differences in abundance driven by a few seemingly random tDRs (Figure S5B; Data S2). These results are consistent with studies showing a limited pool of sRNAs meeting the criteria of miRNA in *A. fumigatus* (Ozkan et al. 2017; Kelani et al. 2023; Jochl et al. 2008), and instead more RNAs derived from structural RNAs (e.g., tRNAs and rRNA).

2.3 | tDR Biogenesis Is Primarily Independent of the RNAi Machinery in *A. fumigatus*

The identification of abundant tDRs in our sRNA-seq dataset coupled with previous reports of tRH production in *A. fumigatus* during starvation (Jochl et al. 2008) led us to assess the tDR fraction in more detail. In agreement, we observed an increase in the relative abundance of nuclear-encoded 5'-tRHs from conidia to older mycelium; however, we also detected a large abundance of tRFs, likely generated by cleavage of either mature or precursor tRNA (e.g., by RNase T2). For nuclear-encoded tRFs, we noticed that the relative abundance of tRF-5 in the small RNA fraction increases from conidia to 24- and 48-h-old mycelium where nutrients are more limited, while the tRF-3 fraction decreased across those samples (Figure 2D). However, within mitochondrial tDRs, reads mapped mostly to tRF-3, 5'-tRHs, and tRF-5 (Figure 2D).

Next, we created heat maps of normalized abundance for the tDRs of each tRNA in the wild type fungus (Figure S6, Data S3) using the unitas software package as previously described (Isakova et al. 2020). We observed large differences in the relative abundance of tDRs in each morphotype, suggesting differential stability of different fragments or potentially poor ligation of select tDRs during sequencing library preparation. We saw slight, statistically significant differences in ratios of tDR abundance for some of the RNAi knockout strains (Figure S7, Data S3), especially $\Delta ppdA/B$, as compared to the wild type. In conidia, $\Delta ppdA/B$ appeared to produce fewer tRF-3 derivatives from many tRNAs. Surprisingly, we saw minimal differences in the production of tDRs in the $\Delta dclA/B$ strain. These results suggest that the RNAi machinery, particularly the Dicer-like proteins, do not play a major role in tRNA fragment production in *A. fumigatus* under standard growth conditions.

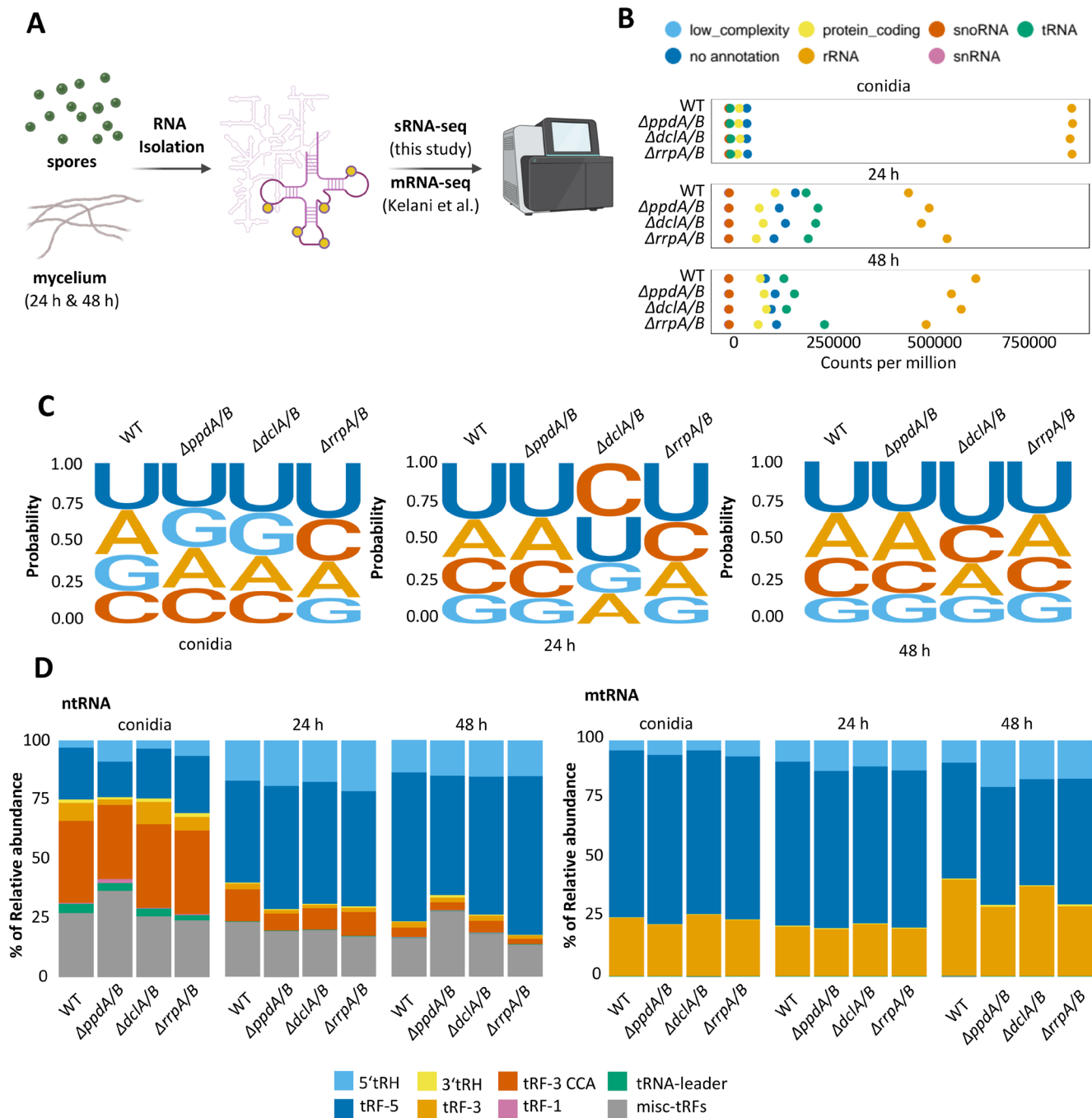


FIGURE 2 | Small RNA abundance in *A. fumigatus* conidia and mycelium. (A) Strains and conditions profiled in this study. The wild-type (CEA17 $\DeltaakuB^{ku80}$) and three knockout strains ($\Delta ppdA/B$, $\Delta dclA/B$, $\Delta rrpA/B$) were collected ($n=3$ biological replicates per condition). RNA samples were collected from fungal conidia and hyphae (24 and 48 h) and Illumina sequenced. Created in BioRender. Blango, M. (2025) <https://BioRender.com/r33p545>. (B) Coverage of small ncRNA types within the profiled samples. Small ncRNAs were annotated with low complexity, no annotation, protein coding, rRNA, snoRNA, snRNA, and tRNA (including nuclear tRNA and mitochondrial tRNA). (C) The distribution of the 5' nucleotide for the respective strains from sRNA-seq data across conidia, 24- and 48-h mycelium. (D) Comparison of tDR relative abundance across samples, and between wild-type and RNAi double knockouts for nuclear-encoded tRNA. $n=3$ biological replicates per condition.

2.4 | tDR-Seq Reveals Altered tDR Patterns Between Fungal Morphotypes

The abundance of tDRs in our sRNA-seq, together with the known difficulties of sequencing tRNAs using standard methodologies (e.g., poor adapter ligation and decreased reverse transcription efficiency), led us to perform a tRF & tRNA-seq (here

denoted tDR-seq) approach, where RNA ends are repaired and common methylation modifications are removed to facilitate ligation and readthrough, respectively (Figure 3A; Data S4). We focused on the wild-type fungal morphotypes for simplicity as only minor differences were observed between the RNAi knockouts and wild type in the previous analysis. Isolation of RNA again confirmed that the proportion of small RNAs increases

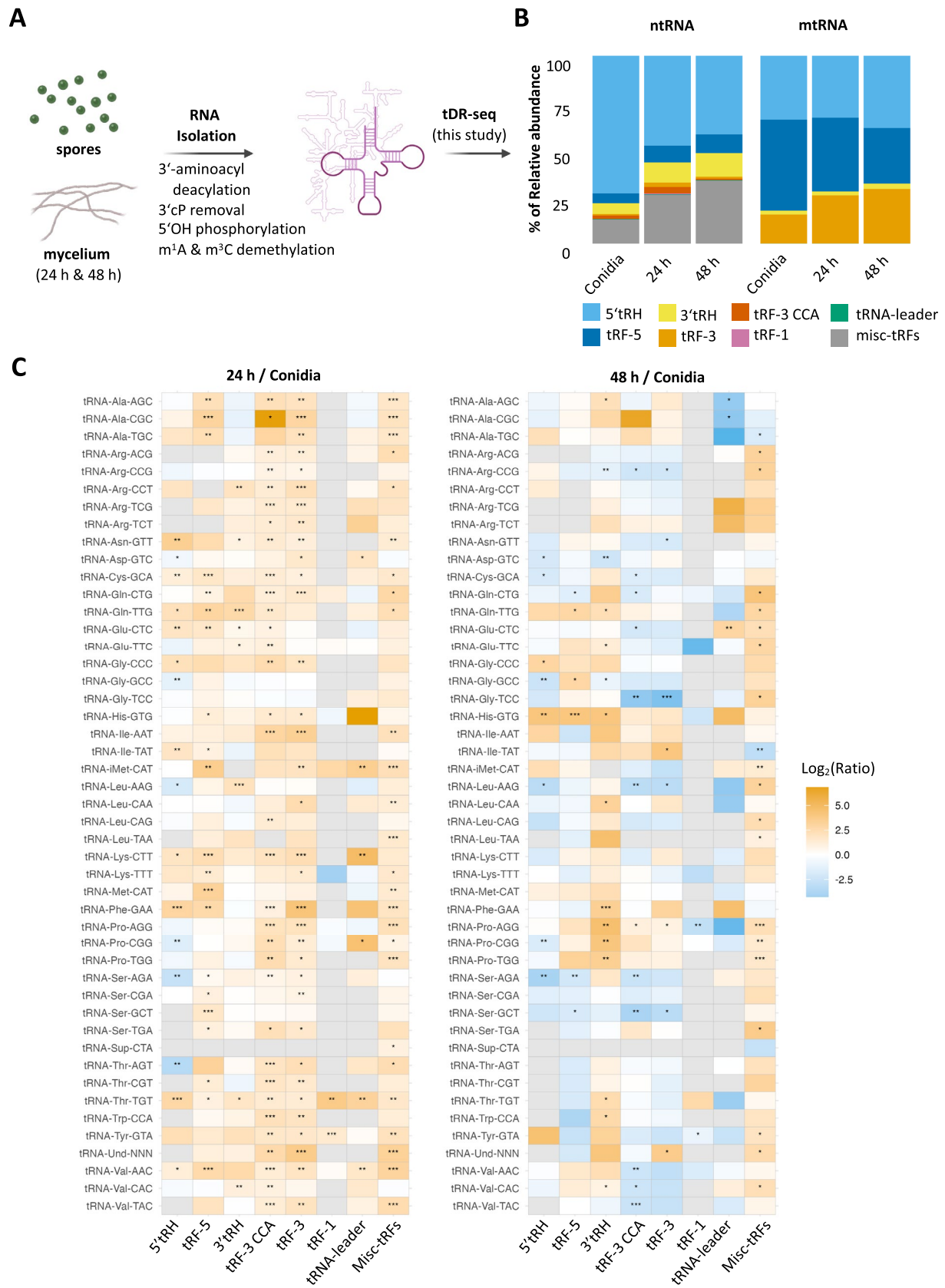


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at 48-h compared to 24-h mycelium and conidia (Figure S8A). After unitas analysis of tDR-seq, a larger relative proportion of reads mapped to rRNA fragments (rRFs) than in the sRNA-seq;

however, a closer look at the distribution of tDRs indicated a higher proportion of tRHs compared to the sRNA-seq analysis for the nuclear-encoded tRNAs (Figure S8B; Data S5). As

FIGURE 3 | Dynamics of tDR expression in *A. fumigatus* conidia and mycelium revealed by tDR-seq. (A) RNA samples ($n = 3$) were collected from wild-type *A. fumigatus* conidia and hyphae. Samples were pre-treated to remove specific modifications and resolve tDR ends prior to library preparation. RNA molecules of up to 50 nt were selected for tDR-seq using Illumina sequencing. Created in BioRender. Blango, M. (2025) <https://BioRender.com/e54f907>. (B) Relative abundance of different tDRs across samples for tDRs derived from nuclear- (ntRNA) and mitochondria-encoded (mtRNA) tRNAs. (C) Heatmap showing ratio of abundance of nuclear-encoded tDR species and their tRNA of origin (with reads collapsed into a single isodecoder) for mycelium and conidia timepoints from tDR-seq dataset. Plots show ratio of counts per million mapped reads based on fractionated counts produced with unitas v. 1.7.0 (Gebert et al. 2017) for mycelium (24-h or 48-h) versus conidia. *Denotes significant differences as defined in the methods; $n = 3$ biological replicates per condition.

compared to the tDR distribution from sRNA-seq (Figure 2D), tDR-seq revealed more 5' and 3' tRHs relative to other tDRs, consistent with an approach that fixes RNA ends prior to library preparation (Figure 3B). Interestingly, the relative abundance of 5' tRHs with nuclear origin was much higher in the conidia (~75%) after the tDR-seq approach (Figure 3B). Further analysis revealed that a majority (~73%) of tDRs identified in the conidia came from a particular Asp(GTC) nuclear-encoded tRNA, with the 5' tRH being the most abundant fragment (Data S5). The length of the tRNA fragments was more consistent with tRHs for the nuclear-encoded tRNAs, but the mitochondria-encoded tDRs had a wider range, with a relative enrichment of smaller fragments in hyphae (Figure S8C), consistent with the literature (Isakova et al. 2020).

A heat map of tDR abundances for each tRNA again revealed substantial differences in abundance and fragment distribution between tRNAs with different or even the same isoacceptor (Figure S9, Data S5). Even though many of the detected tRHs derive from Asp(GTC), other tRNAs like His(GTG) and Pro(CGG) also produce a relatively high relative abundance of tRHs. Many of the nuclear-encoded tRNAs appeared to have stable tRF-3 CCA fragments, with the previously identified Tyr(GTA)-tRF-3 CCA being among the most abundant fragments (Figure S9; Kelani et al. 2023). The counts per million reads of tDRs generally increased in 24-h mycelium relative to conidia, as demonstrated by tDR ratios between the morphotypes (Figure 3C). Collectively, these results corroborate that the sRNA-seq data that the *A. fumigatus* small transcriptome consists mostly of sRNAs derived from structural RNAs (e.g., rRNAs and tRNAs) under standard growth, with different relative abundances in the small RNA fraction depending on the morphotype.

2.5 | dsRNA Overexpression Limits *A. fumigatus* Growth Independent of Dicer-Like Proteins

Our observation that the small RNAs produced from the *pksP* IRT had no predicted off-target sites aside from the target *pksP* mRNA led us to hypothesize that high levels of dsRNA may itself be detrimental to *A. fumigatus*, perhaps through competition for substrates with endogenous pathways. In fact, *A. fumigatus* has 10 annotated dsRBPs (FungiDB.org; Alvarez-Jarreta et al. 2024), with 5 being putative RNase III enzymes (InterPro). These contain two Dicer-like proteins (DclA, AFUB_059350; DclB, AFUB_100200; Table S1), a mitochondrial RNase III enzyme that is an ortholog of MRPL3 from *N. crassa* (AFUB_012200; Lee et al. 2010), the nucleolar RNase III enzyme Rnt1 (AFUB_053000), and a small RNase III domain-containing protein of unknown function found

only in *A. fumigatus* and a few close relatives (AFUB_054390) (Figure S10A). The five other dsRBPs are less well-studied, but in several cases have roles in rRNA assembly or function (Ribosomal Protein S5, AFUB_088030; 37S Ribosomal Protein S5, AFUB_059110; RAD52 ortholog, AFUB_064040; AFUB_031820; and AFUB_067420). Two of these dsRBPs are predicted to reside in the mitochondria (MRPL3; 37S Ribosomal Protein S5), with the others likely found in the cytoplasm and/or nucleus. Expression profiling from previously published mRNA-seq data on conidia, 24-h mycelium, and 48-h mycelium revealed stage-specific expression of many of these genes often influenced by expression of the core RNAi components (Kelani et al. 2023 and Figure S10B), suggesting that specific subsets of proteins might compete for dsRNA binding at different growth stages.

We tested the contribution of sRNA production to the observed growth inhibitory effect of the *pksP* IRT in liquid culture. As expected, induction of the inverted-repeat transgene in the wild-type strain limited fungal biomass as before (Kelani et al. 2023), but surprisingly, this phenotype was also similar in the $\Delta dclA/B$ strain after induction, albeit less pronounced (Figure 4A). Low expression of dsRNA via leaky TetON induction in the knock-out strain resulted in a mild decrease in growth even in the un-induced RNAi double knockout strain, suggesting that dsRNA may itself be growth inhibitory. Although the untreated culture of the $\Delta dclA/B$ strain was significantly decreased compared to the control strains, there remained a difference between the induced and uninduced condition in the $\Delta dclA/B$ *pksP* IRT strain. As a control, we grew each of these strains in the absence of the inverted-repeat transgene, with no major differences in growth observed for the $\Delta dclA/B$ double knockout strain (Figure S10C). Overexpression of the *pksP* IRT was also inhibitory in the single *dclA* and *dclB* deletion strains, again with a stronger inhibition upon induction with doxycycline compared to the wild type (Figure 4B, Figure S10D), indicating that the observed effect was not due to the pairwise deletion of the dicer-like proteins. Future study will be required to tease apart the complete regulation of dsRNA in the sRNA-limited environment of *A. fumigatus* under laboratory growth conditions.

3 | Discussion

Small ncRNAs are key regulators of gene expression and stress response in many organisms, including fungi. Here, we provide an improved description of the sRNA landscape of *A. fumigatus* by leveraging sRNA-seq and tDR-seq. The pool of *A. fumigatus* tDRs appears to be heterogeneous, dynamic, and primarily independent of the canonical RNAi machinery under standard

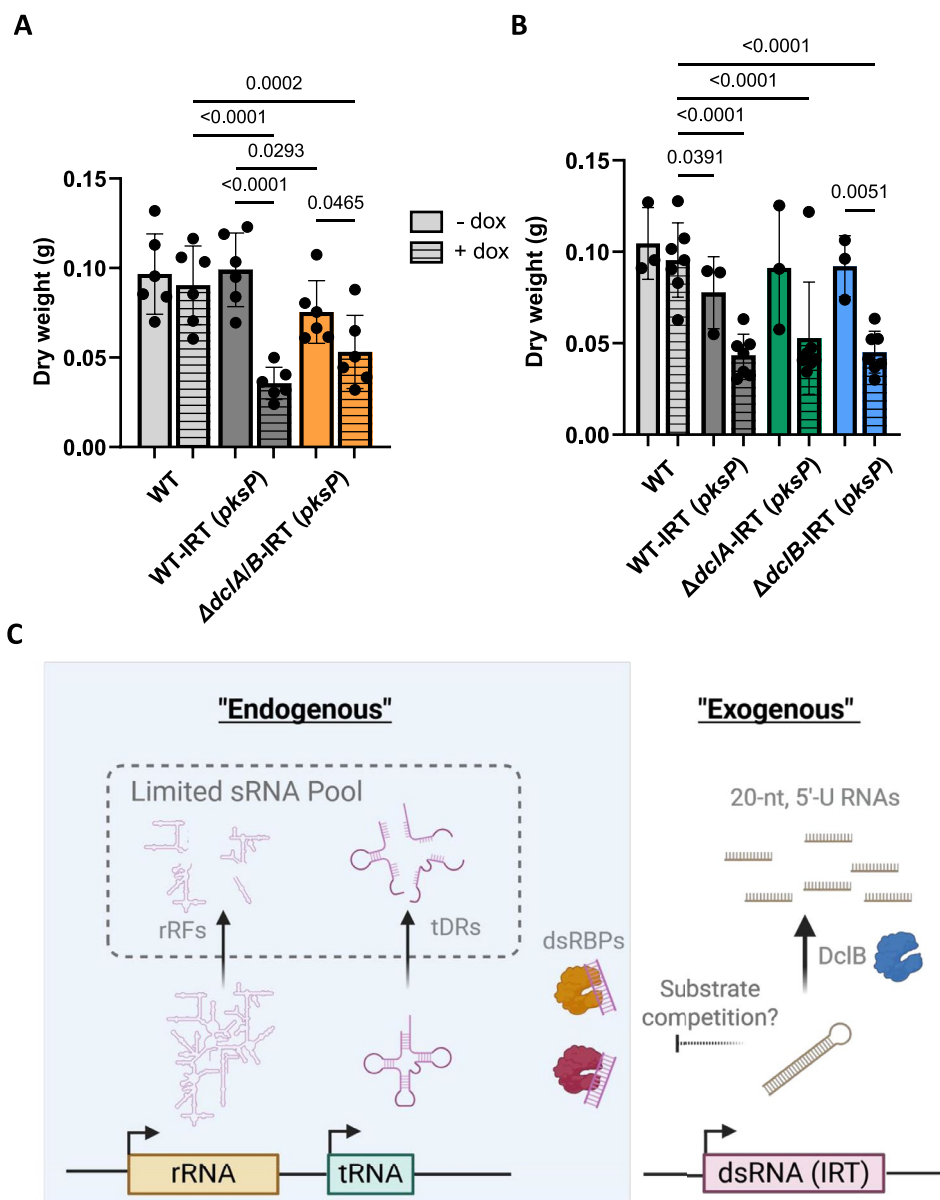


FIGURE 4 | Excess double-stranded RNA induces growth arrest. (A) Measurement of fungal dry weight (biomass) after 24-h growth in liquid culture in the presence or absence of doxycycline as an inducer of *pksP* IRT in the RNAi knockout strains or compared to a wild-type strain. Relevant comparisons were performed with two-way ANOVA and Šidák's multiple comparisons posttest from six biological replicates representing the average of two technical replicates each. (B) Measurement of fungal dry weight (biomass) after 24-h growth in liquid culture in the presence or absence of doxycycline as an inducer of *pksP* IRT in the *dcl* single-knockouts or wild-type strain and a wild-type strain lacking the IRT. Relevant comparisons were performed with two-way ANOVA and Šidák's multiple comparisons posttest from at least three biological replicates representing the average of two technical replicates each. (C) Working model of small RNA landscape in *A. fumigatus*. High levels of exogenous dsRNA can be induced to produce 20-nt 5'-U-containing RNAs and sequester dsRBPs from normal functions to limit growth. Created in BioRender (<https://BioRender.com/r33p545>). Blango, M. (2026) <https://BioRender.com/epon469>.

growth conditions. We took advantage of the limited endogenous sRNA landscape to assess the features of the sRNAs produced by *A. fumigatus* upon induction of an inverted-repeat transgene. We were not surprised to observe an accumulation of 5' U-containing 20-nt sRNAs derived from the *pksP*-IRT transgene, but it was notable that these sRNAs were primarily generated from two consecutive loci in the 3' half of the 500-bp dsRNA. We noted that sRNAs were produced in the wild-type and $\Delta dclA$ IRT strains, but not the $\Delta dclB$ or $\Delta dclA/B$ strains, corroborating our previous finding that DclB is required for

small RNA generation from the IRT in *A. fumigatus*. A second round of sequencing of a previously created *pabA* IRT expressing strain again showed an enrichment of 5'-U-containing 20-nt sRNAs from primarily the 3' end of the transcript, suggesting a conserved feature of hairpin cleavage by DclB.

We performed sRNA-seq to evaluate the dynamics of small ncRNAs during two distinct morphological phases of *A. fumigatus* growth and identified expected classes of small ncRNAs, with reads derived from rRNA and tDRs exhibiting the highest

abundance in conidia and mycelium. Although a potential sign of RNA degradation, our previous experiences with *A. fumigatus* in combination with our quality control steps and the literature (Jochl et al. 2008), suggest that rRNA- and tRNA-derived RNAs do accumulate over the lifecycle, particularly at later growth timepoints (e.g., 48-h mycelium). It remains possible that artifacts could be introduced by the RNA isolation method (Kim et al. 2012), a feature which should be more systematically explored in the future. Interestingly, the abundance of tDRs observed in our sRNA-seq and tDR-seq did not appear to correlate with our previous measurement of the overall tRNA abundances from mycelium under the same growth conditions using Nano-tRNAseq, where for example, Ser(GCT) was found to be the most abundant tRNA isoacceptor (Bruch et al. 2025). This supports the hypothesis that tDR biogenesis is a regulated process in *A. fumigatus*, and not solely a product of RNA degradation (Jochl et al. 2008).

Some previous studies have indicated that RNAi components like Dicer play a role in tDR biogenesis (Babiarz et al. 2008; Cole et al. 2009; Haussecker et al. 2010), but our results suggest that *A. fumigatus* Dicer-like proteins play a minor role under standard growth conditions. As compared to wild type, we did observe a slight decrease in the production of some tDRs in the conidia of $\Delta ppaA/B$ and an increase at later growth stages (i.e., 48-h-old hyphae). Argonaute proteins are known to stabilize tRFs in some systems (Kuscu et al. 2018), and it remains possible that the total pool of mature tRNAs susceptible to cleavage is lower in the $\Delta ppaA/B$ strain, based on the downregulation of genes associated with transcription observed previously (Kelani et al. 2023). It is worth noting that we did not observe loss of sRNAs in the $\Delta ppaA/B$ *pksP* IRT sRNA-seq experiment (Figure S2D), suggesting that the argonautes may not play a major role in sRNA stabilization in *A. fumigatus*. Further study will be required to assign the enzymes responsible for fragment generation.

To better assess the tDR repertoire of *A. fumigatus*, we performed tDR-seq (Zhang et al. 2021), which revealed an enriched pool of tRHs compared to sRNA-seq consistent with the literature (Jochl et al. 2008). The tDR-seq dataset showed an abundance of rRNA reads, a smaller fraction of tDRs, and other RNAs consistent with previous work (Gustafsson et al. 2025). We were not surprised to observe an increase in tRH detection using tDR-seq compared to sRNA-seq (Figure S8), but it was somewhat surprising how strong this enrichment was for conidia compared to the mycelium conditions. It will be interesting in the future to more accurately compare the end status of RNAs from each of these conditions, as this result may suggest interesting biochemistry in the conidia, perhaps for longer term protection of the RNAs.

We next assessed the ability of overexpression of the inverted-repeat transgene to induce growth arrest in liquid culture as previously observed in (Kelani et al. 2023), but now in the absence of dicer-like protein components. We were surprised to see that growth was limited, even when the RNAi system was disrupted by deletion of the dicer-like proteins. The growth disruption in the $\Delta dclA/B$ and $\Delta dclB$ strains incapable of producing sRNAs (Figure 4A,B) suggested that the growth disruption is independent of sRNA production and that off-target effects of

the sRNAs are not the major driver of growth arrest. Instead, we hypothesized that the dsRNA itself may serve as a signal for growth arrest. In this case, high levels of dsRNA may compete with dsRBPs for endogenous substrates and impede normal biology, akin to an actively replicating retrotransposable element (Figure 4C). Such a mechanism could offer a natural brake on growth in response to abnormally high levels of dsRNA and give an advantage to unaffected cells, but this remains speculative. Additional work using complementation with catalytically inactive RNAi machinery will be required to fully understand the mechanism(s) of action. Of note, previous work showed that overexpression of a single-stranded mRNA in similar growth conditions does not lead to growth inhibition (Shopova et al. 2020), providing additional support for a potential inhibitory role of dsRNA itself.

In conclusion, our study (1) reveals DclB-dependent production of canonical 20-nt, 5' U-containing small RNAs by *A. fumigatus* upon dsRNA induction; (2) further defines the dynamics of the limited pool of endogenous small RNAs consisting largely of rRNA and tRNA-derived RNAs using sRNA- and tDR-seq; and (3) provides an intriguing hypothesis for a link between dsRNA metabolism and growth arrest in *A. fumigatus* independent of sRNA production that should be investigated further in the future.

4 | Experimental Procedures

4.1 | Strains, Culture Conditions, and Growth Assays

Throughout this study, *A. fumigatus* strains (Table S2) were cultivated on *Aspergillus* minimal medium (AMM) agar plates and liquid culture as described previously (Kelani et al. 2023). It is important to note that the strain termed wild type here lacks nonhomologous end joining, which may have unintended impacts on the biology observed. For measuring the dry weight, *A. fumigatus* strains were cultivated on AMM-agar plates for 5 days, then 1×10^8 spores were added to 50-mL AMM liquid culture. Strains with inverted repeat elements were supplemented with 10 μ g/mL of doxycycline. Flasks with AMM liquid culture were shaken at 200 rpm at 37°C in the dark for 24 h. Mycelium was collected through Miracloth (Millipore) and dried in the oven at 65°C for 7 days prior to measurement of the dry weight. For the sRNA-seq of inverted-repeat transgene strains, 1×10^8 freshly harvested spores were inoculated into 50 mL AMM and incubated at 37°C and 200 rpm. After 16 h, 10 μ g/mL of doxycycline was added to the denoted inverted-repeat transgene strains, followed by 8 h of additional growth (total of 24 h).

The DclB complementation plasmid was created using primers defined in Table S3 (plasmid map in Figure S3). Fungal protoplasts were created of the $\Delta dclB$ +IRT (*pksP*) strain and transformed with linearized plasmid according to the protocol described in (Kelani et al. 2023). Colonies were selected for integration of the pyrithiamine resistance gene (*ptrA*) in the safe haven 2 (SH2 region; Pham et al. 2020). After three restreaks, DNA was collected and southern blots performed according to (Kelani et al. 2023), using restriction enzymes *NruI* and *NheI*

and flanking sequence as probe. Positive clones were confirmed in growth assays after ~40 h by assessing for the presence of silencing after complementation of *dclB* (white colonies indicate functional silencing of *pksP* gene).

4.2 | RNA Isolation for IRT-sRNA-Seq

For IRT-sRNA-seq, RNA was isolated as in (Kelani et al. 2023) with several improvements. Collected mycelium was first transferred to a 2-mL screw-cap tube filled 1/3 full with 0.5 mm soda lime glass beads (BioSpec). One milliliter of TRIzol (Thermo Fischer Scientific, Dreieich, Germany) was added to efficiently lyse the cells using the FastPrep-24 Homogenizer by subjecting the cells to three rounds of 30 s, 4.0 m/s homogenization (1 min break between repetitions) and 5 min of incubation at room temperature. Two hundred microliters of chloroform was added to each sample, vigorously mixed and incubated for 3 min at room temperature before centrifuging the samples at 12,000×g for 15 min at 4°C. The upper aqueous phase containing the RNA was carefully transferred to a clean RNase-free tube and an equal volume of pure ice-cold ethanol was added and mixed thoroughly. Purification of RNA was carried out by using the RNA Clean and Concentrator kit (Zymo Research GmbH, Freiburg, Germany) following the manufacturer's instructions. To eliminate DNase contamination, total RNA was treated with DNase I, RNase-free (Thermo Fisher Scientific, Darmstadt, Germany) at 1 U per µg of RNA. Following a 1-h incubation at 37°C, the reaction was terminated by adding EDTA (5 mM final concentration) at 65°C for 10 min and total RNA was collected using the RNA Clean and Concentrator-25 kit (Zymo Research) following the manufacturer's instructions. DNase-free RNA concentrations were measured with the Qubit RNA BR Assay kit (Thermo Fisher Scientific, Darmstadt, Germany). Purified RNA was quantified using the Qubit RNA BR Assay kit (Thermo Fisher Scientific, Darmstadt, Germany) on the Qubit Flex Fluorometer.

4.3 | RNA Isolation for sRNA- and tDR-Seq

RNA isolation was performed as described previously (Kelani et al. 2023). Briefly, fungal mycelium was harvested from liquid culture using Miracloth (Millipore) and then disrupted in liquid nitrogen with a precooled mortar and pestle. Approximately 0.5 g of homogenized mycelium was transferred to a 2-mL Eppendorf tube. To this, 800 µL of TRIzol was added and vigorously vortexed. The tubes were briefly frozen in liquid nitrogen and then allowed to thaw on ice. Following thawing, 160 µL of chloroform was added to the mixture, vortexed, and centrifuged for 5 min at 4°C at maximum speed. The resulting aqueous upper phase was carefully transferred to a fresh 2-mL tube without disturbing the interphase. RNA extraction from the aqueous phase involved the addition of 1 volume of phenol/chloroform/isoamyl alcohol (25:24:1, v/v), followed by brief vortexing and centrifugation for 5 min at 4°C. This extraction process was repeated until no interphase was visible, and an additional extraction with 400 µL chloroform was performed. RNA was precipitated by adding 400 µL isopropanol for 20 min, followed by centrifugation for 20 min at 4°C. The resulting pellet was washed with 700 µL

70% ethanol and air-dried at 37°C for 5 min before being resuspended in RNase-free water. To remove any remaining DNA, a DNase treatment was carried out using 2 units of TURBO DNase (Thermo Fisher) per 10 µg RNA for 30 min at 37°C in a total volume of 100 µL. The resulting DNase-free RNA was then processed as described above.

4.4 | Library Preparation and Sequencing

sRNA-seq libraries were prepared by Novogene after RNA quality assessment (Figures S2A–C and S4A) using the TruSeq Small RNA Library Preparation Kit according to the manufacturer's protocol, then sequenced using the NovaSeq 6000 instrument in a single-read, 50-bp based mode at 10 million reads of sequencing depth. For tDR-seq performed by CD Genomics (tRF&tRNA-seq), the following treatments for modifications were performed: (1) 3'-aminoacyl (charged) deacylation to 3'-OH for 3' adaptor ligation, (2) 3'-cP removal to 3'-OH for 3' adaptor ligation, (3) 5'-OH (hydroxyl group) phosphorylation to 5'-P for 5'-adaptor ligation, and (4) m¹A and m³C demethylation for efficient reverse transcription. Finally, RNA libraries were sequenced using an Illumina HiSeq 2500 instrument in a 50-bp single-read mode at 10 million reads of sequencing depth.

4.5 | Data Analysis

TrimGalore version 0.6.8 (<https://github.com/FelixKrueger/TrimGalore>) was used for quality checks and removing adaptors of sequencing reads. SortMeRNA version 4.3.7 (Kopylova et al. 2012) was used to remove potential rRNA reads, with the rRNA database generated from Rfam (<https://rfam.org/>) and silva (<https://www.arb-silva.de/>). In silico removal of rRNA reads did not alter analysis results and was subsequently omitted for downstream analyses. Sequencing data were analyzed with unitas v. 1.7.0 (Gebert et al. 2017) using default parameters and *A. fumigatus* A1163 ASM15014v1 as the reference genome. *A. fumigatus* nuclear tRNAs were referenced from GtRNAdb (Thornlow et al. 2020), mitochondrial tRNAs were scanned from the mitochondrial genome (Joardar et al. 2012) using the 27 tRNAs predicted by tRNAscan-SE (Chan et al. 2021). All analyses were performed using "R" version 4.3.1 (2023-06-16) and Rstudio (2022-02-02), an integrated development environment (IDE) for R or Prism 10.4.0 software (GraphPad Software). Figures were made with the ggplot2 package (version 3.4.2) in R. Graphs for read coverage were created using the SparK package (Kurtenbach and Harbour 2019). *P*-values were determined by ANOVA or two-tailed Student's *t*-tests where appropriate. For the tRNA fragment heatmaps, pairwise comparisons were performed using linear mixed-effects models (LMMs) with sample as a fixed effect and replicate as a random effect. When the LMMs were singular or failed to converge, fixed effects linear models were employed instead. We computed post hoc pairwise contrasts using the *emmeans* package (Lenth 2025), and controlled for multiple comparisons using the Benjamini-Hochberg method. In all cases, differences between the groups were considered significant at a *p*-value of < 0.05. **p*-value of < 0.05, **< 0.01, ***< 0.001.

Author Contributions

Xiaoqing Pan: conceptualization, experimental work, data analysis and interpretation, writing – original draft, writing – review and editing. **Abdulrahman A. Kelani:** conceptualization, experimental work, data analysis and interpretation, writing – original draft, writing – review and editing. **Lukas Schrettenbrunner:** experimental work. **Swatika Prabakar:** experimental work. **Bhawana Israni:** experimental work. **Matthew G. Blango:** conceptualisation, funding acquisition, data analysis and interpretation, writing – original draft, writing – review and editing.

Acknowledgments

We would like to thank Pamela Lehenberger for excellent technical assistance and Axel Brakhage, Alexander Bruch, and Xiuqiang Chen for helpful discussions. The work presented here was generously supported by the “Bundesministerium für Forschung, Technologie und Raumfahrt” (BMFT; Federal Ministry of Research, Technology and Space) Germany, Project FKZ 01K12012 “RFIN – RNA-Biologie von Pilzinfektionen”. Additional funding support came from the ‘Deutsche Forschungsgemeinschaft’ (DFG; German Research Foundation) under Germany’s Excellence Strategy – EXC 2051 – Project-ID 390713860. The funders had no role in the design of the study. ChatGPT (GPT-4 model, OpenAI) was used to search for relevant software packages and coding functions.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The sRNA-seq data are available under NCBI Gene Expression Omnibus (GEO) identifier GSE267454. These data were collected simultaneously with previously published mRNA-seq data (Kelani et al. 2023) available using identifier GSE223618. The tDR-seq data are available under identifier GSE267453. The sRNA-seq for the inverted-repeat transgene overexpression is available using GSE286229 and GSE300894.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** mmi70042-sup-0001-AppendixS1.

pdf. **Data S1:** mmi70042-sup-0002-DataS1.xlsx. **Data S2:** mmi70042-sup-0003-DataS2.xlsx. **Data S3:** mmi70042-sup-0004-DataS3.xlsx. **Data S4:** mmi70042-sup-0005-DataS4.xlsx. **Data S5:** mmi70042-sup-0006-DataS5.xlsx.